

Welfare and Spending Effects of Consumption Stimulus Policies

March 25, 2026

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Abstract

Using a heterogeneous agent model calibrated to match spending dynamics over four years following an income shock (?), we assess the effectiveness of three fiscal stimulus policies implemented during recent recessions. Unemployment insurance (UI) extensions are the “bang for the buck” winner when the metric is effectiveness in boosting utility. Stimulus checks are second-best and have two advantages (over UI): they arrive faster, and are scalable. A temporary (two-year) cut in wage taxation is considerably less effective than the other policies and has negligible effects in the version of our model without a multiplier.

Keywords stimulus checks, unemployment insurance extensions, payroll tax cuts, HANK/heterogeneous agent models, marginal propensity to consume, spending multipliers

JEL codes E21, E62, H31

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Replication Package: [10.5281/zenodo.17861977](https://zenodo.org/record/105281)

Other Materials:

Jupyter Notebook: <https://econ-ark.org/materials?hafiscal>
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Forthcoming, *Quantitative Economics*

The views expressed in this paper are those of the authors and do not necessarily represent those of the Federal Reserve Board, the Deutsche Bundesbank and the Eurosystem, or Statistics Norway. This project has received funding from the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation programme (grant agreement No. 851891) and from the Research Council of Norway (grant No. 326419).

The authors would like to thank seminar participants at the Board of Governors of the Federal Reserve System, Deutsche Bundesbank, the European Central Bank, the Norwegian University of Science and Technology, Oslo Macro Group, the Society for Computational Economics in 2023 and 2024, Statistics Norway, the University of Pennsylvania, Vanderbilt University, Johns Hopkins University, and West Virginia University for valuable feedback. The authors would also like to thank Sasha Indarte, Adrian Monninger, Plutarchos Sakellaris, three anonymous referees, and the editor Morten Ravn for insightful comments on earlier versions of the paper. Thanks also to the <https://github.com/econ-ark/econ-ark.org> team members who helped with the project, particularly Alan Lujan, Akshay Shanker, and Matthew White (alphabetically).

1 Introduction

Although many empirical studies argue that potential and existing entrepreneurs face borrowing constraints, so far there has been little work on how these constraints affect aggregate capital accumulation and wealth inequality through entrepreneurial choices. Do these financial constraints hamper aggregate capital accumulation and, if so, how big is this effect? What effect do these constraints have on wealth inequality: do they exacerbate it or mitigate it? These are potentially important forces to understand the consequences of policy reforms that affect the tightness of these borrowing constraints, such as changes in the leniency of bankruptcy laws and in the degree of enforcement of property rights.

In this paper we analyze the role of borrowing constraints as determinants of entrepreneurial decisions (entry, continuation, investment, and saving), and their effects on wealth inequality and aggregate capital accumulation, in a framework that matches the observed wealth inequality very closely. In the presence of borrowing constraints, the decision to invest, the fraction of entrepreneurs, and the size distribution of firms depend on the distribution of assets in the economy. Because of this interaction, it is key to perform such an analysis in a model that matches well the extreme concentration of wealth observed in the data.

We find that more restrictive borrowing constraints generate less inequality in wealth holdings but also reduce average firm size, the number of people engaging in entrepreneurial activities, and aggregate capital accumulation. Our results also indicate that voluntary bequests are an important channel allowing some high-ability workers to establish or enlarge an entrepreneurial activity. If there were only accidental bequests, there would be fewer very large firms and less aggregate capital, but also less wealth concentration.

These findings are based on a quantitative life cycle model with altruism across generations and entrepreneurial choice, in an environment in which debt repayment cannot be perfectly enforced. The amount that entrepreneurs can borrow depends on their observable characteristics, and the entrepreneurs' assets act as collateral for their debts. Since the implicit rate of return for entrepreneurs is higher than the rate for workers, entrepreneurs have a higher saving rate, which is consistent with the data. We calibrate the parameters of the model to match key moments of the data and discuss the implications of the model and its components for entrepreneurial choice and wealth inequality. We show that our model with entrepreneurial choice matches very well the observed distribution of wealth, for both entrepreneurs and nonentrepreneurs.

Section 2 first documents the relationship between wealth and entrepreneurship and then surveys the evidence that entrepreneurs are borrowing constrained. Section 3 describes the model and our calibration procedure. Section 5 discusses the role of entrepreneurship and voluntary bequests in generating large wealth concentration and studies the aggregate effects of changing the borrowing constraints. Section 6 inspects further the mechanisms at work in our model and compares their observable implications to those in the observed data. Section 7 presents conclusions.

2 Wealth, Entrepreneurship, and Borrowing Constraints

2.1 Which Are the Rich Households?

Wealth holdings are massively concentrated in the hands of a small fraction of households, and this wealth concentration is much larger than the one documented for labor earnings and total income. This observation raises the question of which saving motives generate the amplification in the concentration of wealth with respect to the one in income.

When one is looking at the data, it is clear that there is a tight relationship between being an “entrepreneur” and being rich. We document this relationship using different definitions of entrepreneurship, and we then discuss alternative ways of acquiring wealth.

The SCF asks several questions that we can use to classify a household by its occupational status:

1. “Do you work for someone else, are you self-employed, or what?”
2. “Do you (and your family living here) own or share ownership in any privately held businesses, farms, professional practices or partnerships?”
3. “Do you (or anyone in your family living here) have an active management role in any of these businesses?”

Table 1 Percentage of Entrepreneurs (According to Various Definitions) in the Population and Corresponding Share of Total Wealth Held

	Percent in Population	Share of Total Wealth
Business owners or self-employed	16.7	52.9
All business owners	13.3	48.8
Active business owners	11.5	41.6
All self-employed	11.1	39.0
Self-employed business owners	7.6	33.0

The key message of this table is that, regardless of the specific definition of entrepreneurship used, entrepreneurs are a relatively small fraction of the population and hold a large fraction of the total net worth.

Table 2 documents wealth concentration in the United States: the households in the top 1 percent of the wealth distribution hold about 30 percent of total net worth, and those in the top 5 percent hold more than half of the total.

A whopping 81 percent of those who belong to the top 1 percent of the wealth distribution declare that they either are self-employed or are business owners. All business owners are 76 percent of the richest 1 percent of households.

Regardless of the specific definition of entrepreneurship, entrepreneurs are much richer than nonentrepreneurs. The overall message is that most rich people are entrepreneurs.

Table 2 U.S. Wealth Distribution

	Fraction of People, Top			
	1%	5%	10%	20%
Total net worth held	30%	54%	67%	81%

Table 3 Fraction (%) of Entrepreneurs in a Given Wealth Percentile of the Overall U.S. Wealth Distribution

	Wealth Percentile, Top			
	1%	5%	10%	20%
Business owners or self-employed	81	68	54	39
All business owners	76	62	49	36
Active business owners	65	51	42	30
Self-employed	62	47	38	26
Self-employed business owners	54	39	32	22

2.2 Entrepreneurship and Borrowing Constraints

To estimate the severity of borrowing constraints on entrepreneurial entry and continuation decisions, one would want to know how much potential and existing entrepreneurs would like to borrow, at what interest rate, and how much they are actually able to borrow. Unfortunately, such data are not available.

Many papers have used a variety of data sets and methodologies to indirectly estimate the severity of borrowing constraints for entrepreneurs. Among these works, [Evans and Jovanovic \(1989\)](#) and [Buera \(2006\)](#) estimate structural models of entrepreneurship and find evidence of borrowing constraints; [Gentry and Hubbard \(2004\)](#) and [Eisfeldt and Rampini \(2005\)](#) also argue that costly external financing has important implications for investment and saving decisions. [Hurst and Lusardi \(2004\)](#) have disputed the relevance of borrowing constraints to entrepreneurial entry: they estimate that the probability of entering entrepreneurship as a function of initial wealth is first flat over a large range of the wealth distribution and then increases for the richest workers. We will show that a model of entrepreneurial choice with borrowing constraints is capable of generating this type of entry probabilities as a function of one's own wealth.

The need to accumulate assets in the presence of borrowing constraints may also generate high saving rates among entrepreneurs (or households planning to become entrepreneurs). Using different data sets, [Quadrini \(1999\)](#) and [Gentry and Hubbard \(2004\)](#) show higher saving rates for entrepreneurs than for the rest of the population. Among the self-employed business owners, 29 percent declare that they currently use their own personal assets as collateral to finance their business. Among the self-employed business owners, 18 percent report that they have been turned down for credit, and 9 percent state that they thought of applying but changed their mind because they thought they might be turned down.

3 The Model

3.1 Demographics

We adopt a life cycle model with intergenerational altruism. Households go through two stages of life, young and old age. A young person faces a constant probability of aging during each period $(1 - \pi_y)$, and an old person faces a constant probability of dying during each period $(1 - \pi_o)$. When an old person dies, his offspring enters the model, carrying the assets bequeathed to him by the parent. Appropriately parameterized, this framework generates households for which the average lengths of the working period and the retirement period are realistic. Our model period is one year.

There is a continuum of households of measure one. The households are subject to idiosyncratic shocks, but there is no aggregate uncertainty, as in Bewley (1977).

3.2 Preferences

The household's utility from consumption is given by

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}. \quad (1)$$

The households discount the future at rate β , and, in addition, they discount the utility of their offspring at rate η . In the purely life cycle version ($\eta = 0$), individuals put no weight on the utility of their descendants. In the perfectly altruistic version ($\eta = 1$), individuals care about their descendants as much as themselves. We assume exogenous labor supply.

3.3 Technology

Each person possesses two types of ability: entrepreneurial ability (θ), the capacity to invest capital more or less productively, and working ability (y), the capacity to produce income out of labor.

Entrepreneurs can borrow and invest capital in a technology whose return depends on their own entrepreneurial ability. When the entrepreneur invests k , the production is given by θk^ν , where $\nu \in [0, 1]$. Entrepreneurs thus face decreasing returns from investment, since their managerial skills become gradually stretched over larger projects (as in Lucas (1978)).

Workers can save (but not borrow) at a riskless, constant rate of return. The nonentrepreneurial sector is represented by a standard Cobb-Douglas production function:

$$F(K_c, L_c) = AK_c^\alpha L_c^{1-\alpha}, \quad (2)$$

where K_c and L_c are the total capital and labor inputs in the nonentrepreneurial sector and A is a constant. In both sectors, capital depreciates at a rate δ .

3.4 Credit Market Constraints

As in Kehoe and Levine (1993) and Cooley, Marimon, and Quadrini (2004), the borrowing constraints are endogenously determined in equilibrium and stem from the assumption that contracts are imperfectly enforceable.

We assume that the entrepreneurs who borrow can either invest the money and repay their debt at the end of the period or run away without investing it and be workers for one period. In the latter case, they retain a fraction f of their working capital k (which includes their own assets and borrowed money), and their creditors seize the rest. Hence, the entrepreneur's assets act as collateral, although the loan need not be fully collateralized.

3.5 Households

Each young individual starts the period with assets a , entrepreneurial ability θ , and worker ability y and chooses whether to be an entrepreneur or a worker. The young's value function is

$$V(a, y, \theta) = \max \{V_e(a, y, \theta), V_w(a, y, \theta)\}, \quad (3)$$

where $V_e(a, y, \theta)$ is the value function of a young entrepreneur:

$$V_e(a, y, \theta) = \max_{c, k, a'} \{u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1 - \pi_y)EW(a', \theta')\}, \quad (4)$$

$$a' = (1 - \delta)k + \theta k^\nu - (1 + r)(k - a) - c, \quad (5)$$

$$u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1 - \pi_y)EW(a', \theta') \geq V_w(f \cdot k, y, \theta), \quad (6)$$

$$a \geq 0. \quad (7)$$

The enforcement constraint (6) ensures that the entrepreneur prefers repaying to defaulting: in case of default, the entrepreneur retains fraction f of working capital and becomes a worker.

The value function for the young worker is

$$V_w(a, y, \theta) = \max_{c, a'} \{u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1 - \pi_y)W_r(a')\} \quad (8)$$

subject to $a' = (1 + r)a + (1 - \tau)wy - c$ and $a \geq 0$, where w is the wage and τ is a proportional payroll tax financing old-age social security.

The old entrepreneur can choose to continue the entrepreneurial activity or to retire:

$$W(a, \theta) = \max \{W_e(a, \theta), W_r(a)\}. \quad (9)$$

3.6 Equilibrium

Let $\mathbf{x} = (a, y, \theta, s)$ be the state vector for an individual, where s distinguishes young workers, young entrepreneurs, old entrepreneurs, and old retired. A stationary equilibrium is given by a risk-free interest rate r , wage rate w , and tax rate τ ; allocations $c(\mathbf{x})$,

$a(\mathbf{x})$, occupational choices, and investments $k(\mathbf{x})$; and a constant distribution $m^*(\mathbf{x})$, such that:

- The functions c , a , and k solve the maximization problems described above.
- The capital and labor markets clear.
- The wage and interest rates are given by the marginal products of each factor of production.
- The social security budget constraint is balanced period by period.
- The distribution m^* is the invariant distribution for the economy.

4 Parameterizing the Model

The empirical definition of entrepreneurship that we use for the calibration must be consistent with the notion of entrepreneur in our framework. We use the SCF data to classify as entrepreneurs the households that declare that they are self-employed, that they do own a business, and that they have an active management role in it. This definition eliminates managers (who are not likely to think of themselves as self-employed) and the business owners who do not manage the business that they own.

Our general calibration strategy is to reduce the number of parameters that we use to match the data as much as possible. We divide our parameters into two sets. The first set either can be easily estimated from the data without using our model or has been estimated by many previous studies. We use the second set of parameters to match some relevant moments of the data.

Panel A of the table shows the set of parameters that we take from other studies. We take the coefficient of relative risk aversion to be 1.5, a value close to those estimated by Attanasio, Banks, Meghir, and Weber (1999). The depreciation rate δ is 6 percent. The share of income that goes to capital in the nonentrepreneurial sector is 0.33. The probabilities of aging and of death are such that the average length of the working life is 45 years, and the average length of the retirement period is 11 years. The logarithm of the income process y for working people follows an AR(1) with persistence 0.95, as estimated by Storesletten, Telmer, and Yaron (2004). The variance is chosen to match the Gini coefficient for earnings of 0.38. The social security replacement rate is 40 percent of average income. In the baseline case we set $\eta = 1$ (perfect altruism).

Panel B lists the remaining parameters: β , the vector $\boldsymbol{\theta}$, $\mathbf{P}_\theta$, ν , and f . We consider a very parsimonious calibration and allow for only two values of entrepreneurial ability: zero and a positive number. We use these six parameters to pin down the following moments: the capital-output ratio, the fraction of entrepreneurs in the population, the fraction of entrepreneurs exiting entrepreneurship, the fraction of workers becoming entrepreneurs, the ratio of median net worth of entrepreneurs to that of workers, and the wealth Gini coefficient.

Table 4 Parameters of the Model

		Value	Source(s)
<i>Panel A: Fixed Parameters</i>			
σ	Risk aversion	1.5	Attanasio, Banks, Meghir, and Weber (1999)
δ	Depreciation rate	.06	Stokey and Rebelo (1995)
α	Capital share	.33	Gollin (2002)
A	TFP scaling	1.0	Normalization
π_y	Prob. staying young	.978	See text
π_o	Prob. staying old	.911	See text
$\mathbf{P}_y$	Income transition	See text	Storesletten, Telmer, and Yaron (2004)
p	Social security	40% avg. income	Kotlikoff et al. (1999)
η	Altruism	1.0	Perfect altruism
<i>Panel B: Calibrated Parameters</i>			
β	Discount factor	.865	
θ	Entrepreneurial ability	[0, .51]	
$\mathbf{P}_\theta$	Ability transition	See text	
ν	Decreasing returns	.88	
f	Default retention	75%	

Given the features matched in the calibration, we analyze how well the model matches the overall distribution of wealth and the distributions of wealth for entrepreneurs and workers. We use the implications of the model in this respect as a check of the validity of our model.

5 Results

We first study the two versions of our model (one without and one with entrepreneurs) and discuss their ability to reproduce the observed inequality in wealth.

Table 5 Comparing Data and Models with and without Entrepreneurs

	Capital– Output	Wealth Gini	Entre- preneurs	Percentage Wealth in Top			
				1%	5%	20%	40%
U.S. data	3.0	.80	7.55%	30	54	81	94
Model w/o entrepreneurs	3.0	.60	0%	4	20	58	95
Baseline w/ entrepreneurs	3.0	.80	7.50%	31	60	83	94

5.1 The Model without Entrepreneurs

The model economy without entrepreneurs assigns zero entrepreneurial ability to everyone. This model economy produces a distribution of wealth that is much less concentrated than that in the data and, in particular, does not explain the emergence of the large estates that characterize the upper tail of the distribution of wealth.

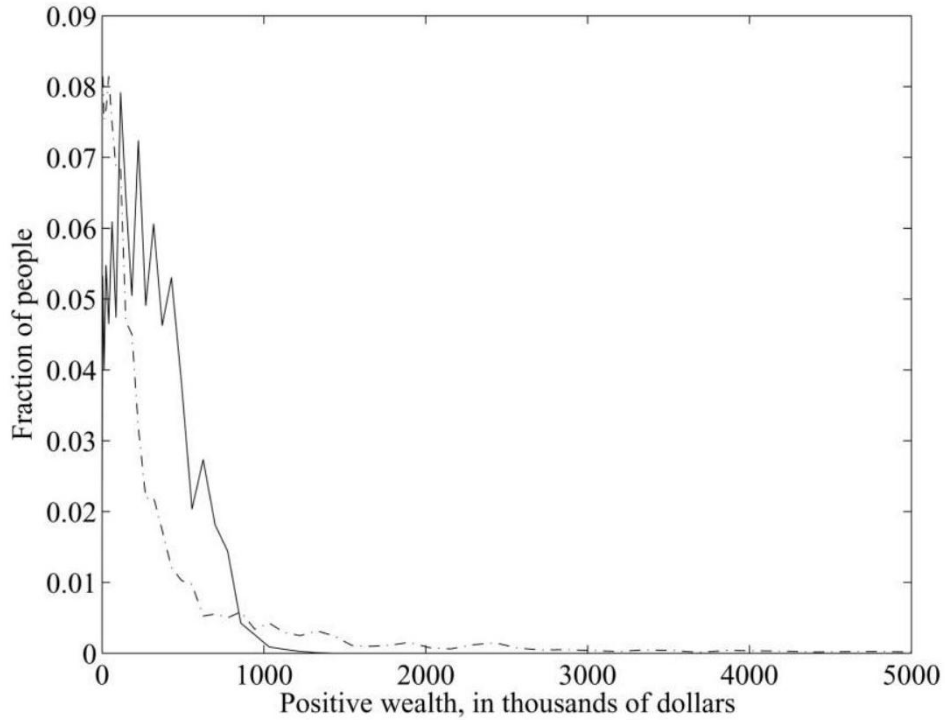


Figure 1 Distribution of wealth, conditional on wealth being positive, for the whole population. Dash-dot line: data; solid line: model without entrepreneurs.

While the data on wealth display a fat tail, in the model without entrepreneurial choice, all households hold less than \$1.1 million.

5.2 The Model with Entrepreneurs

In our baseline simulation the equilibrium interest rate r is 6.5 percent; the share of total wealth held by entrepreneurs is 29 percent, compared with 33 percent in the data; and the degree of decreasing returns to the entrepreneurial technology is 0.88. This parameterization matches the distribution of wealth very well both for the overall population and for the subpopulation of entrepreneurs.

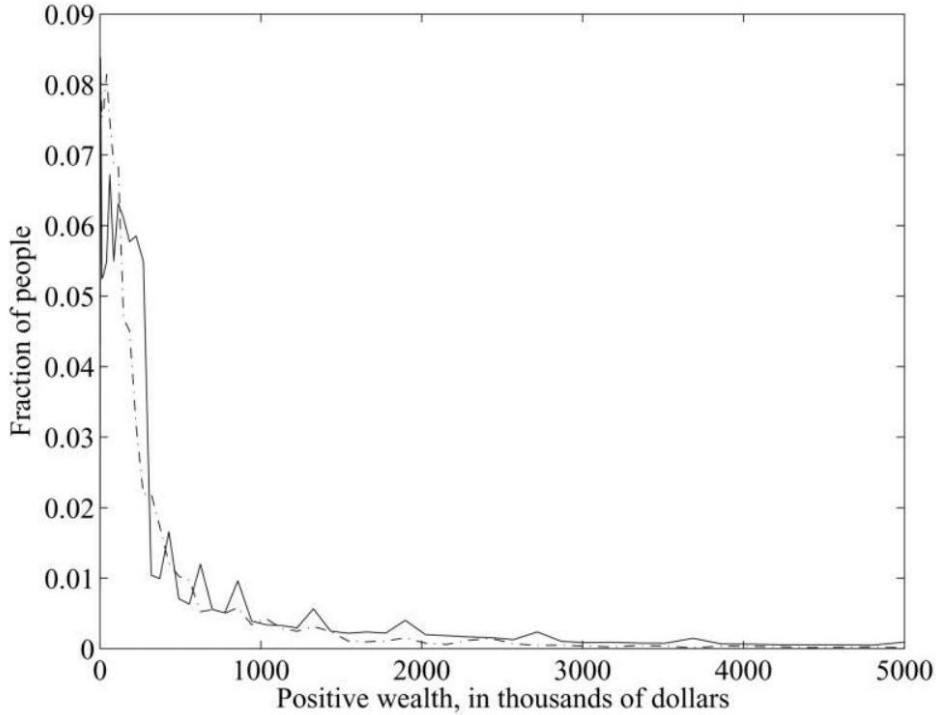


Figure 2 Distribution of wealth, conditional on wealth being positive, for the whole population. Dash-dot line: data; solid line: baseline model with entrepreneurs.

The distribution of wealth for the population of entrepreneurs displays a much fatter tail than the one for workers. Contrary to the model without entrepreneurial choice, the baseline model generates distributions of wealth for both entrepreneurs and nonentrepreneurs with a significant mass of people who have more than \$1.1 million. The nonentrepreneurs in the right tail of the wealth distribution are former entrepreneurs or descendants of entrepreneurs.

Those with low entrepreneurial ability (who are thus workers) exhibit buffer stock saving behavior highlighted by [Carroll \(1997\)](#): if their assets are low, they save; if high enough, they dissave. The saving rate of those with high entrepreneurial ability and enough assets to become entrepreneurs is positive and considerably higher than that for workers. The combination of higher returns together with the budget constraint generates a very high saving rate for entrepreneurs.

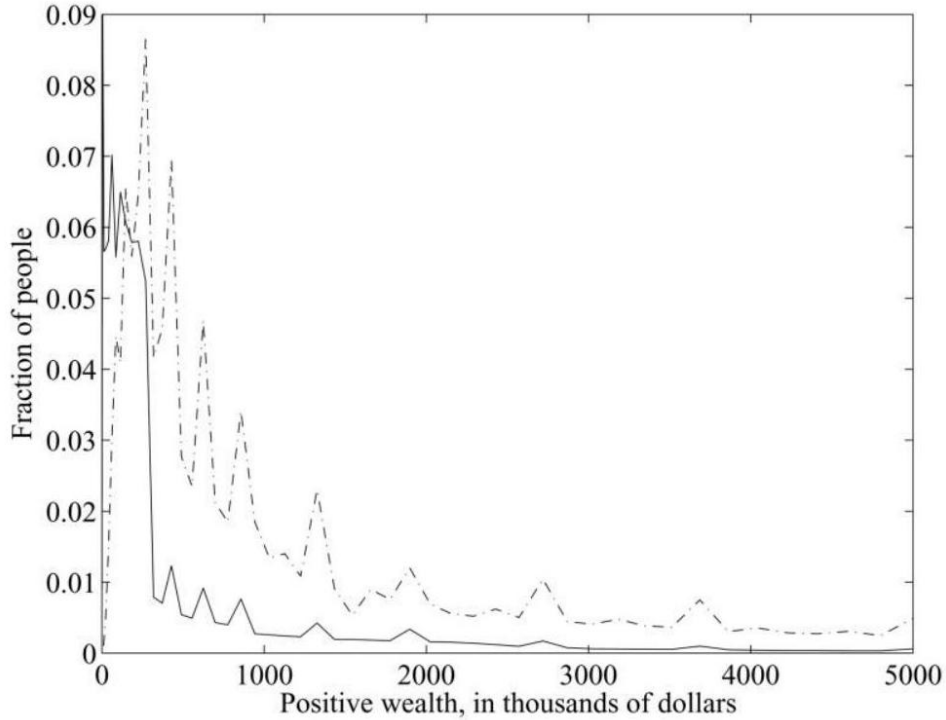


Figure 3 Distribution of wealth, conditional on wealth being positive, in the baseline model with entrepreneurs. Solid line: workers; dash-dot line: entrepreneurs.

Borrowing constraints generate a large amount of heterogeneity in the firm size distribution. The tail is generated by the entrepreneurs who have remained in business for a long period and have thus had time to save and increase the size of their firms.

5.3 The Borrowing Constraints

We examine the effect of changing the tightness of the borrowing constraints by increasing f , the fraction of working capital that cannot be seized by creditors, from 0.75 to 0.85.

The capital-output ratio drops drastically, from 3.0 to 2.7, and the fraction of entrepreneurs falls from 7.5 percent to 6.9 percent since fewer high-ability individuals can borrow and start a firm.

5.4 Bequests

In the baseline economy households are altruistic toward their offspring. We study what happens when households do not care about their descendants and all bequests are accidental.

The absence of the voluntary bequest motive reduces the incentives to accumulate capital and run larger firms. The total capital of the economy decreases from 3.0 to 2.5,

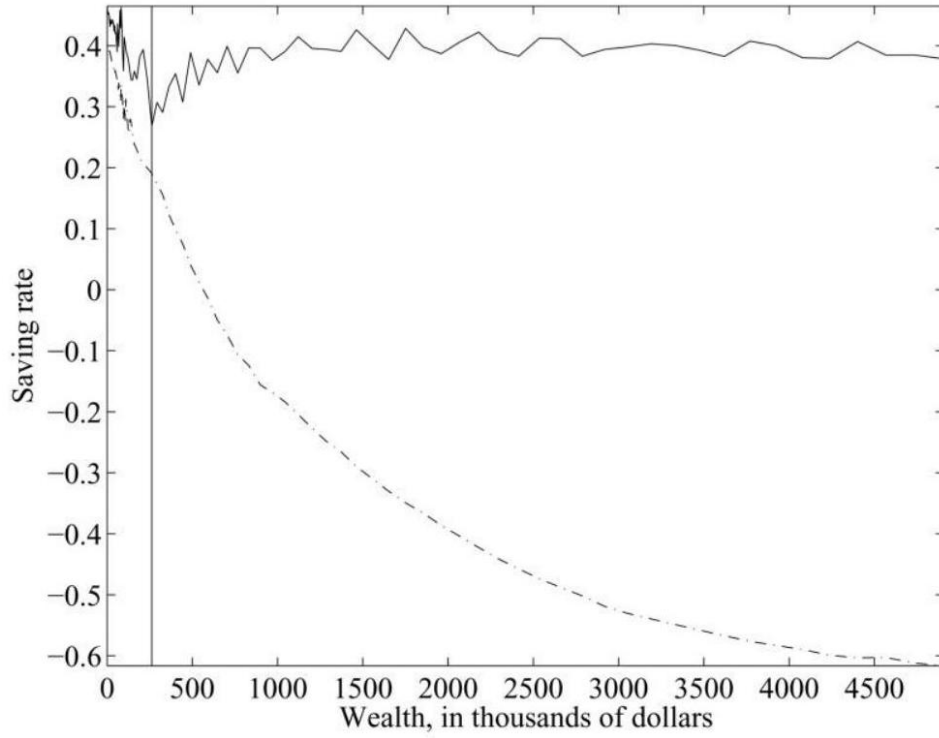


Figure 4 Saving rate for highest-ability workers. Solid line: those with high entrepreneurial ability; dash-dot line: those with no entrepreneurial ability; vertical line: asset level at which high-entrepreneurial ability individuals enter entrepreneurship.

Table 6 The Role of Borrowing Constraints and Voluntary Bequests

	Capital- Output	Interest Rate	Wealth Gini	Entre- preneurs	% Wealth in Top			
					1%	5%	20%	40%
U.S. data	3.0	...	.80	7.55%	30	54	81	94
Baseline	3.0	6.5%	.80	7.50%	31	60	83	94
More stringent ($f = .85$)	2.7	7.5%	.70	6.90%	24	49	75	91
No altruism ($\eta = 0$)	2.5	9.3%	.70	7.55%	21	45	73	90
$\eta = 0$, recalib. $\beta = .88$	3.0	6.4%	.80	7.90%	28	57	81	94

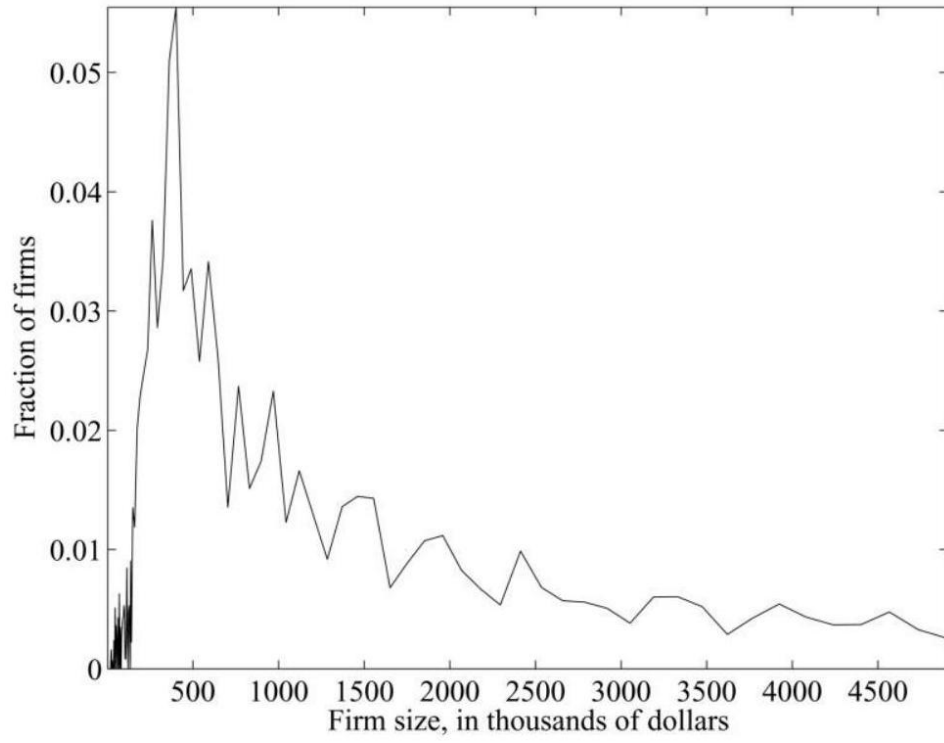


Figure 5 Firm size distribution, baseline model with entrepreneurs.

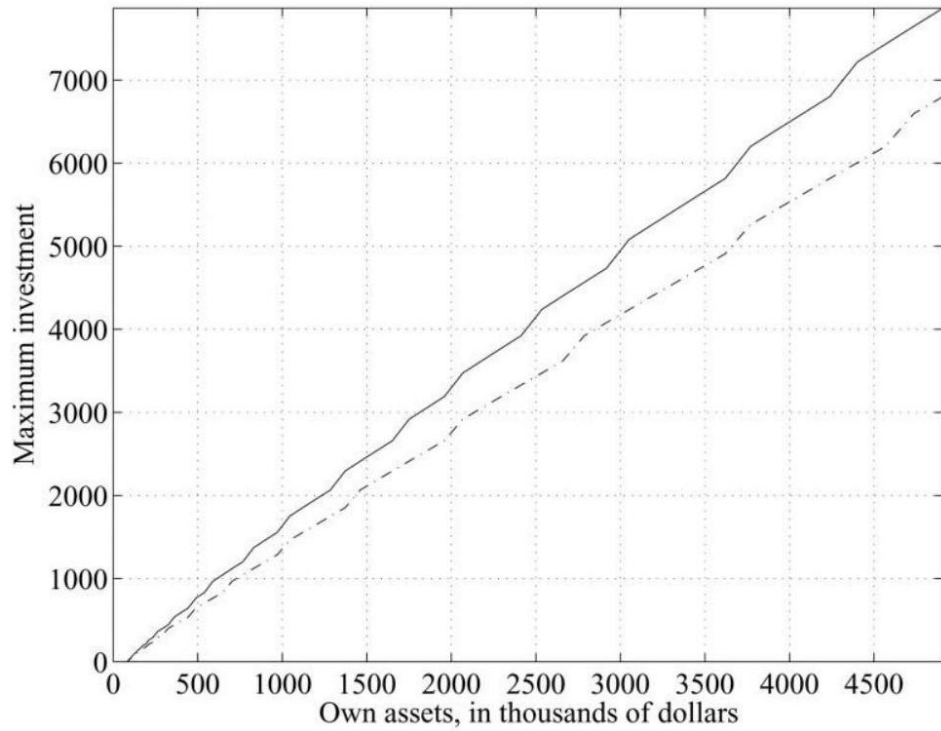


Figure 6 Maximum investment. Solid line: baseline; dash-dot line: more restrictive borrowing constraints.

and the concentration of wealth drops substantially. As also shown by De Nardi (2004), voluntary bequests are fundamental in explaining the concentration of wealth.

5.5 Impulse responses

(Placeholder for future impulse response analysis.)

6 Inspecting the Model’s Mechanisms

Recent literature has cast doubt on the relevance of borrowing constraints to entrepreneurial entry (Hurst and Lusardi, 2004) and on the size of the returns to entrepreneurship (Moskowitz and Vissing-Jørgensen, 2002). High wealth inequality in our model is generated by the combination of occupational choice in the presence of borrowing constraints and high potential returns to entrepreneurship. We check here whether the observable implications generated by our model are consistent with the observed data.

6.1 Borrowing Constraints and Entry

The main finding of Hurst and Lusardi (2004) is that the probability of entering entrepreneurship is almost flat over a large portion of the wealth distribution and then increases for the richest workers. There are two main points worth discussing.

First, the results of our paper do not depend in important ways on the fact that the financial constraint affects the decision to become an entrepreneur. Rather, it is the greater incentive to save after entry that is crucial for the results.

Second, our model of occupational choice with borrowing constraints produces entry decisions that are consistent with the Hurst–Lusardi finding. We generate many samples of households from our model and estimate a probit regression of entry probability as a function of wealth.

Both Hurst and Lusardi’s estimates and our confidence intervals are consistent with an entry probability that is a convex function of wealth. In the presence of borrowing constraints, a worker with high entrepreneurial ability is likely to save to enter entrepreneurship. As a result, we are not very likely to observe many high-ability potential entrepreneurs among the poor. Given only one type of entrepreneurial ability and that none of the calibrated parameters were chosen to match this aspect of the data, it is remarkable how our model is not inconsistent with flat entry probabilities over large sections of wealth holdings.

6.2 Returns from Private Business Ownership

Moskowitz and Vissing-Jørgensen (2002) cast doubt on the assumption that entrepreneurs face potentially high rates of return. We compare the cross-sectional

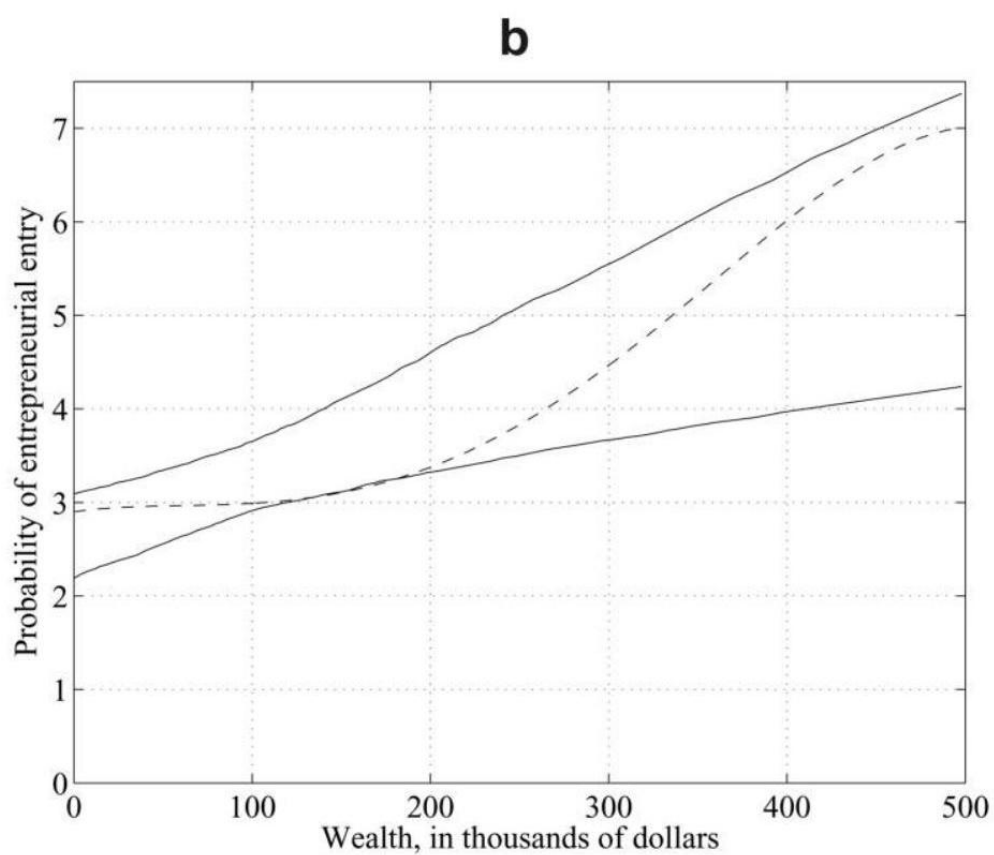
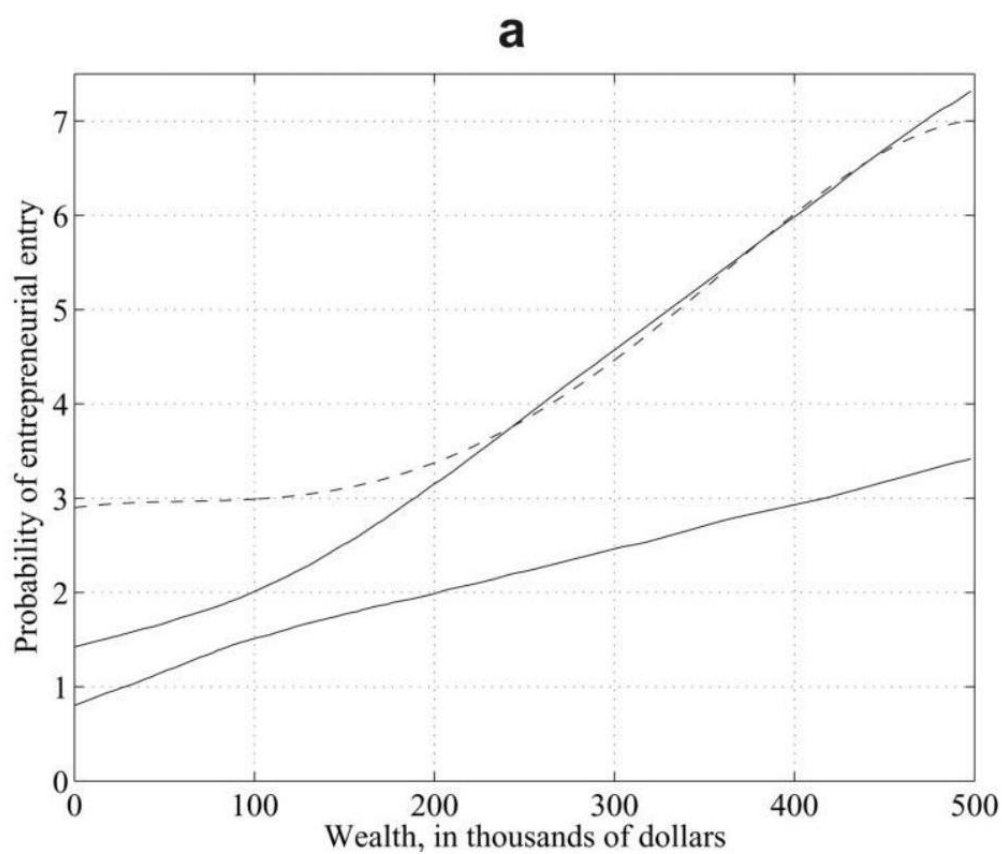


Figure 7 Probability of entering entrepreneurship as a function of own wealth as estimated by Hurst and Lusardi (dashed line), and confidence interval generated by two versions of the model (solid lines). *a*, Benchmark model. *b*, Benchmark with a small fraction of nonentrepreneurial self-employed.

distribution of returns to entrepreneurship in our model and in the data using the 1989 SCF wave.

Table 7 Distribution of Rates of Return (%) for Self-Employed Business Owners

	Percentile			
	25th	50th	75th	90th
Only income from business	0	3	25	143
Including wages and salaries	10	40	125	520

The size of the returns to private equity crucially hinges on how income from the firm is divided between entrepreneurial wages and return to capital. The median distribution of returns is 49 percent in our model compared to 40 percent in the data, so our median entrepreneurial return is only a little higher in our model than in the SCF data.

7 Conclusions

We developed and solved numerically a model of occupational choice, wealth accumulation, and bequests in which entrepreneurs face an endogenous borrowing constraint that limits the amount that they can borrow. The entrepreneur’s wealth acts as collateral, so the richer the entrepreneur, the higher the amount that he can borrow.

A very parsimonious parameterization of our model generates a wealth distribution that matches the one observed in the data, both for entrepreneurs and for workers. It also produces returns from entrepreneurship and households’ entry probabilities into entrepreneurship as a function of one’s wealth, which are consistent with the ones measured in the data. None of the parameters of the model were chosen to obtain these results.

The key mechanism is that many entrepreneurs face potentially high rates of return but are constrained in the amount that they can borrow. To expand their firm, these entrepreneurs keep saving. In doing so, they become richer and richer. The most successful dynasties share their fortunes with their children, some of whom will keep the family firm going, thus expanding the dynasty’s fortune even more.

We show that the tightness of borrowing constraints and voluntary bequests are main forces in determining the number of entrepreneurs, the size of their firms, the overall wealth concentration in the population, and the aggregate capital accumulation.

These results have implications for policy analysis, such as subsidized loans to entrepreneurs and estate taxes. Subsidized loans would make it cheaper for the entrepreneurs to borrow but would also change their incentives to default, making the effects of this policy a priori ambiguous. Taxing bequests may decrease inequality, while at the same time reducing the amount of entrepreneurial wealth that could be used as collateral, and thus may affect both the number of entrepreneurs and the total capital of the economy, as shown by Cagetti and De Nardi (2004).

Appendices

A Income and Entrepreneurial Ability Processes

We assume that the income process is lognormal and AR(1). We approximate it with a five-point discrete Markov chain, using the method described in Tauchen and Hussey (1991).

The resulting grid points for the vector $\mathbf{y}$ for the income process (normalized to an average of one) are

$$[0.2468 \quad 0.4473 \quad 0.7654 \quad 1.3097 \quad 2.3742],$$

and the transition matrix $\mathbf{P}_y$ is

$$\mathbf{P}_y = \begin{bmatrix} .7376 & .2473 & .0150 & .0002 & .0000 \\ .1947 & .5555 & .2328 & .0169 & .0001 \\ .0113 & .2221 & .5333 & .2221 & .0113 \\ .0001 & .0169 & .2328 & .5555 & .1947 \\ .0000 & .0002 & .0150 & .2473 & .7376 \end{bmatrix}.$$

We assume that the entrepreneurial ability process is uncorrelated with the income process. The two values for ability θ are zero (no entrepreneurial ability) and a positive value (0.514), and the transition matrix $\mathbf{P}_\theta$ is

$$\mathbf{P}_\theta = \begin{bmatrix} .964 & .036 \\ .206 & .794 \end{bmatrix}.$$

A.1 Correlation between Abilities

We have assumed that working ability (y) and entrepreneurial ability (θ) are uncorrelated. One important piece of evidence in favor of this specification is that we replicate fairly well the income of entrepreneurs prior to starting their businesses. Using the PSID, the labor earnings over the previous five years of those who enter entrepreneurship in a given period are 1.32 times the labor income during the previous five years of those who choose to remain workers. Our simulations reproduce this feature: the ratio is 1.35. This correlation arises endogenously because of borrowing constraints—among the high-entrepreneurial ability workers, those who receive high labor earnings realizations can save more and are more likely to enter entrepreneurship.

As a robustness check, we study the effects of allowing for positive correlation of 0.4 between the two ability processes. The resulting wealth distribution is close to the benchmark: the richest 1 percent hold 33 percent of the total wealth. The main difference is that the number of entrepreneurs decreases to 5.2 percent because people with high entrepreneurial ability tend to have a higher option value of remaining in the nonentrepreneurial sector.

B The Algorithm

The algorithm proceeds as follows:

- Construct a grid for the state variables. The maximum asset level is chosen so that it is not binding.
- Fix a tax rate τ , an interest rate r , and a wage rate w . Taking these as given, solve for the value functions using value function iteration.
- Construct the transition matrix $\mathbf{M}$. Compute the associated invariant distribution, starting from a guess for $\boldsymbol{\pi}$ and iterating on $\boldsymbol{\pi}' = \mathbf{M}\boldsymbol{\pi}$ until convergence.
- Compute total savings and total capital invested in the entrepreneurial sector from the invariant distribution.
- Compute r and w implied by the nonentrepreneurial aggregate production function, and iterate until convergence on the factor prices.
- Compute the social security system imbalance and iterate on τ until outlays equal revenues.

The computation of the value functions is nonstandard because of the endogenous borrowing constraints. For each state $\mathbf{x}$, the endogenous constraint specifies a maximum amount $k(\mathbf{x})$ that can be borrowed, but this function depends on the value functions themselves. We initialize $\hat{k}(\mathbf{x}) = k_{\max}$, solve the value functions conditional on this constraint, then update $\hat{k}(\mathbf{x})$ by finding the maximum investment for which the entrepreneur would not default, and iterate until $\hat{k}$ converges.

C Results in a Model Without the Splurge

Full appendix PDF available at: [GitHub: Subfiles/Appendix-NoSplurge.pdf](#)

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C.1 Introduction

C.2 Matching the iMPCs without the splurge

C.3 Estimating discount factor distributions without the splurge

C.4 Multipliers in the absence of the splurge

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APPENDICES

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